BPS Link POS Specification – HYPERCOM

Version Number: 1.9

Provided By

Document Title	BPS Link POS Specification – HYPERCOM
Prepared By	Name: Kittiphong Ngoenthavon Signature: Date:
Reviewed By	Name: Patchara Kalampavanich Signature: Date:
Approved By	Name: Ching Hung Hui Signature: Date:
Remarks	
File Name	BPS Link POS Specification - HYPERCOM.docx

Revision History					
Rev.	Section	Description	Date	Author	Remarks
1.0	All	Initial draft	2019-06-17	Kittiphong N.	
1.1	1	Update ● 1.3 Error Handle Add ● 1.6 Serial Port Configuration ● Appendix j EDC normal workflow	2019-10-30	Kittiphong N.	
1.2	All	Update Specification ● Add link POS parameter for both request and response. ● Add link POS specification for Redemption.	2020-08-17	Kittiphong N.	
1.3	All	Update Specification - Revise Sale request POS to EDC - Revise Sale response EDC to POS - Revise Sale QR request POS to EDC - Revise Sale QR response EDC to POS	2020-09-17	Kittiphong N.	
1.4	1	Update 3.3 response code.	2021-01-15	Kittiphong N.	

1.5		Edit Command - Sale command 20 to 56 - Void - Sale welfare card 56+TAX Add command settlement	2021-01-18	Kittiphong N.	
1.6	7-10	Add Command - Summary Report Command 90 - Detail Report Command 91 - Settlement Command 50	2021-04-01	Anatthiwat L.	
1.7	3, 4, 6, 7, 8, 9, 10	Remove unused fields Add VAT Rebate support to - Sale (All Payment Types) Command 20 - Sale (All Card Scheme Types) Command 56 Add HO field the response of - Settlement Command - Summary Report Command - Detail Report Command	2021-04-02	Patchara K.	
1.8	3, 6, 8, 9, 10, 11	Add HA field request of - Settlement Command - Summary Report Command - Detail Report Command Update 30 field response of QR code.	2021-05-28	Thongchai B.	

1.9	10, 11, 12, 13	Add New command - detail report - all host - summary report แบบ all host Edit format data field type HO – Batch total Add Transaction DATA 3.4 Mapping issuer card index 3.5 Host Name	2021-06-14	Thongchai B.	

Table of Contents

1 PROTOCOL 8

1.1 MESSAGE FROM POS TO TERMINAL 8
1.2 MESSAGE FROM EDC TO POS 9
1.3 ERROR HANDLING 10
1.4 ERROR HANDLING WHEN LINK POS MESSAGE FORMAT ERROR 11
1.5 ERROR HANDLING WHEN HOST REJECT 12
1.6 SERIAL PORT CONFIGURATION 13

2 LINK POS MESSAGE 14

2.1 MESSAGE STRUCTURE 14
2.2 DATA 16
2.2.1 TRANSPORT HEADER 16
2.2.2 PRESENTATION HEADER 16
2.2.3 FIELD DATA 20
2.2.4 FIELD ELEMENT 21
2.2.5 EXTRA FIELD SEPARATOR 23

3 TRANSACTION DATA 24

3.1 TRANSACTION CODE DEFINITION 24
3.2 FIELD TYPE DEFINITION 25
3.3 RESPONSE CODE 28

4 SALE (ALL PAYMENT TYPES – TRANSACTION CODE 20) COMMAND 30

4.1 REQUEST FROM POS TO EDC 30
4.2 RESPONSE FROM EDC TO POS 30

5 VOID (ALL PAYMENT TYPES – TRANSACTION CODE 26) COMMAND 33

5.1 REQUEST FROM POS TO EDC 33
5.2 RESPONSE FROM EDC TO POS 33

6 SALE QR PAYMENT (TRANSACTION CODE 70) 35

6.1 REQUEST FROM POS TO EDC 35
6.2 RESPONSE FROM EDC TO POS 35

7 SALE (ALL CARD SCHEME PAYMENT TYPES – TRANSACTION CODE 56) COMMAND 38

7.1 REQUEST FROM POS TO EDC 38
7.2 RESPONSE FROM EDC TO POS 38

8 SETTLEMENT (TRANSACTION CODE 50) COMMAND EACH HOST 41

8.1 REQUEST FROM POS TO EDC 41
8.2 RESPONSE FROM EDC TO POS 42

9 SETTLEMENT (TRANSACTION CODE 50) COMMAND ALL HOST 44

9.1 REQUEST FROM POS TO EDC 44
9.2 RESPONSE FROM EDC TO POS 44

10 SUMMARY REPORT (TRANSACTION CODE 90) COMMAND EACH HOST 46

10.1 REQUEST FROM POS TO EDC 46
9.2 RESPONSE FROM EDC TO POS 47

11 DETAIL REPORT (TRANSACTION CODE 91) COMMAND 48

11.1 REQUEST FROM POS TO EDC 50
11.2 RESPONSE FROM EDC TO POS 51

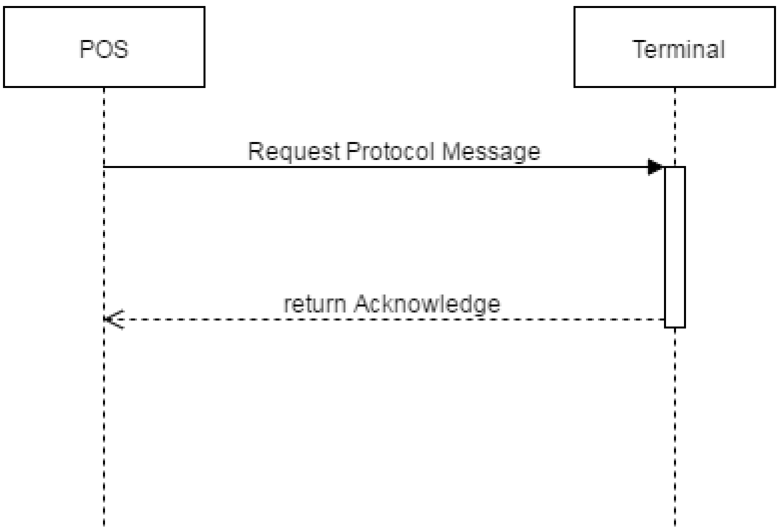
1 Protocol

In order to prevent a single error on the line from causing a message to be lost, a message will be acknowledged when the receiver returns an ACK character.

Only a single unacknowledged message can be outstanding at one time in one direction. The receipt of a message does not imply the previous message sent has been received

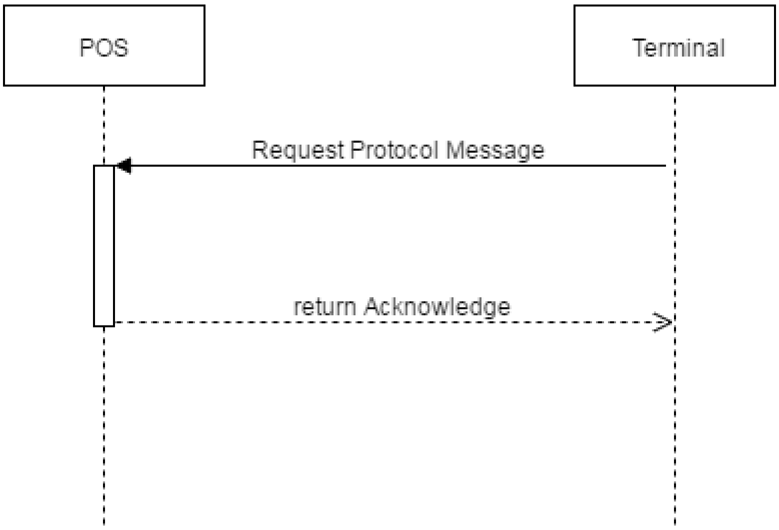
1.1 Message from POS to Terminal

The POS transmits a message. The Terminal acknowledges receipt of the message by transmitting a single ACK (06h) character.



1.2 Message from EDC to POS

The Terminal transmits a message. The POS acknowledges receipt of the message by transmitting a single ACK (06h) character

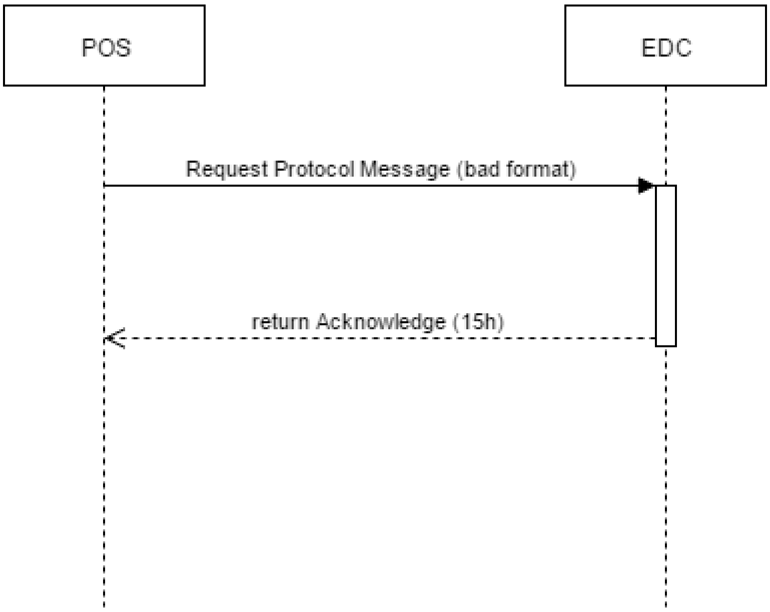


1.3 Error handling

If the POS or the Terminal sends a message, and does not receive the ACK within 5 second, the message should be transmitted again. If the second transmission does not receive an ACK within 5 seconds that message should be treated as undeliverable, and the application should take whatever actions are required to recover.

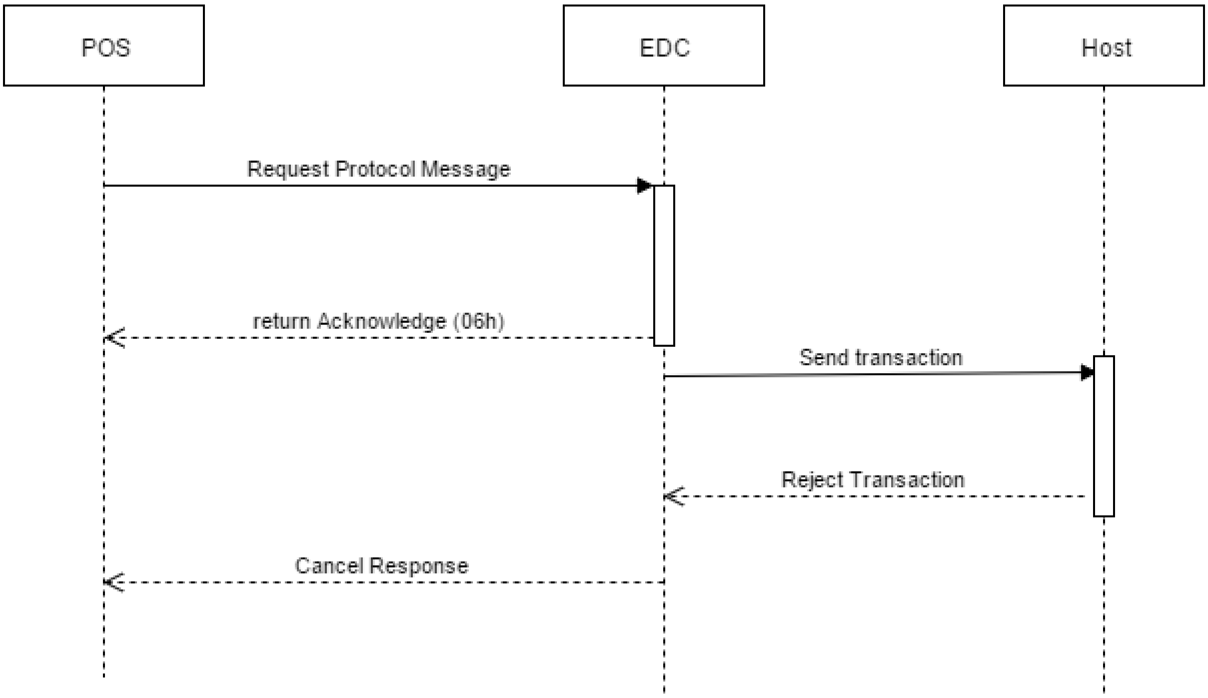
1.4 Error handling when link POS message format error

If the Terminal receives a message in error (Bad Length, missing ETX, incorrect LRC, or unsupported link POS command), terminal sends single acknowledge 15h to notify the message is error. These errors should only be caused by transmission errors, and the retransmission will correct the error. There is no automatic method for recovering from application errors that cause the message to appear corrupted.



1.5 Error Handling when host reject

In case of host rejects transaction, terminal responses cancel message with error reason from host.



1.6 Serial Port Configuration

A POS can connect to an EDC over serial cable or USB (virtual serial) cable with FTDI chip embedded. Both of them can be configured variably to the amount of data we have to exchange between the POS and the EDC.

However, for small content like a payment transaction, we have the default settings on our EDC as following:

- Baud Rate: 9600
- Data Bits: 8
- Stop Bits: 1
- Parity: None

2 Link POS Message

2.1 Message Structure

The messages that are transmitted on the link between the POS and the Terminal will use the following structure

<table>
<tr><td>S
T
X</td><td>LENGTH</td><td colspan="5">DATA</td><td>E
T
X</td><td>L
R
C</td></tr>
<tr><td></td><td></td><td>Transport
Header</td><td>Presentation
Header</td><td colspan="3">Field Data</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Field
Element</td><td>…</td><td>Field
Element</td><td></td><td></td></tr>
</table>

Field	Byte(s)	Value	Description
STX	1	02h	Start of text. This character is used to indicate the start of frame.
LENGTH	2		Length of the following data. This is transmitted in BCD (Binary-Coded Decimal) form. For example, a length of 256 bytes will be transmitted as 02h and 56h. The LLLL field allows the inclusion of binary data in the message. The maximum allowable value for LLL will depend on the implementation.
DATA	Vars		Content of the link POS message

			It consists of a transport header, a presentation header, and field data which is one or more field elements. These difference components are more fully described in the following sections.
ETX	1	03h	End of Text. Logically, this field is not required because of the length indicator (LLLL), but it is included as an extra check that the message was successfully received and that the receiver is in synchronization with the transmitted message.
LRC	1		Longitudinal Redundancy Character. This character is calculated by exclusive OR-ing. Each character following (but not including) the STX up to (and including) the ETX

2.2 Data

2.2.1 Transport Header

The first data in the DATA is the Transport Header. The Transport Header provides flexibility in message routing and allows more than one application to use the same network.

The fields comprising the Transport Header are arranged in the following order.

TRANSPORT HEADER		
Transport Header Type	Transport Destination	Transport Source

Each byte of Field is required to minus 30h to represent the actual value of the field.

For example, 36h minus 30h, the actual value is 06h

Field	Byte(s)	Value	Comment
Transport Header Type	2	‘60’	Used to select the type of application this message is used for.
Transport Destination	4	‘0000’	Allows the POS to select the destination to which this message is to be routed. If this value is ‘0000’, the Terminal will determine the correct destination to route the message to from its internal tables.
Transport Source	4	‘0000’	The first two digits of this field must be ‘00’. The second two digits can be used by the POS for any purpose. Typically, this will be ‘00’ but if there are multiple devices connected to the POS. This field can be used to identify the device the original request originated at.

2.2.2 Presentation Header

The next data in the DATA is the Presentation Header. The Presentation Header indicates if this is a request or a response, the transaction code, a response code, and an indication if there are more messages associated with this message.

The fields comprising the Presentation Header are arranged in the following order.

PRESENTATION HEADER					
Format Version	Request-Response Indicator	Transaction Code	Response Code	More Data Indicator	Field Separator

Each byte of Field (except “Field Separator” 1Ch) is required to minus 30h to represent the actual value of the field.

For example, 36h minus 30h, the actual value is 06h.

Field	Byte(s)	Value	Comment
Format Version	1	‘1’	Indicates what format of the message is supported
Request-Response Indicator	1		‘0’ – This message is a request message which requires a response. ‘1’ – This message is a response message. ‘2’ – This message is a request message which doesn’t require a response.
Transaction Code	2		Identifies the type of transaction the message is preforming. Each byte deduces 30 first, e.g. “32 30”, after deduction, it will be “02 00”.

			See the table in section 3 for a list of valid transaction codes.
Response Code	2		Identifies the result of the transaction in a response. Response code should be set to '00' for all request messages. See the table in section 3 for a list of valid response codes.
More Data Indicator	1		'0' – This is the last message for the transaction. '1' – It has more message come after this message.
Field Separator	1	1Ch	This field is used to make it easier to identify the end of the presentation header when looking at message traces.

2.2.3 Field Data

The Field Data is the most important area for flexibility in the message format.

The method described here allows great benefits, but there is a price to pay for this flexibility; each field element has 2 or 5 bytes of overhead.

The benefits gained from this overhead are:

- Ability to pass binary data.

New features such as Signature Capture will require passing large amounts of data. The overhead required if this cannot be passed as binary data is very high.

- Variable Length Data Fields.
- The data is fairly readable on a message trace.

The Field Data consists of 0 or more Field Elements, arranged one after the other in the message.

FIELD DATA				
Field Element	Field Element	Field Element	…	Field Element

2.2.4 Field Element

The fields comprising the Field Element are arranged in the following order.

FIELD ELEMENT FORMAT			
Field Type	Length	Data	Field Separator

Field	Byte(s)	Value	Comment
Format Type	2		Indicates the type of data that is included in this field element. Each byte is required to minus 30h to represent the actual value of the field. For example; 34h minus 30h, the actual value is 04h. This is an alphanumeric field. Characters 0-9, a-z, and A-Z are available for use.
LLLL	2		Indicates the length in bytes of the data to follow. Length can have a value from 0000 to a value that will not cause the total message to exceed the maximum size allowed for the implementation. Available values for this are detailed in section. LLLL is transmitted in BCD (Binary Coded Decimal) form. The most significant byte is transmitted first, followed by the least significant byte. For example, a length of 256 bytes will be transmitted as 02h 56h.
Data	LLLL		The data for this field. If no data is in this field (for example, when the functionality is indicated by the field type on its own), there will be no data.

Field Separator	1	1Ch	This field is used to make it easier to identify the end of the presentation header when looking at message traces.

2.2.5 Extra Field Separator

Some POS implementations will be easier to develop this interface if field data appears in a particular order, with additional field separators to represent fields that are missing.

To support this, it is acceptable to include extra Field Separator characters (1Ch) between Field Elements.

This can be easily supported in the parsing routine by ignoring any Field Separator characters when it is expecting the first character of the Field Type.

3 Transaction Data

3.1 Transaction Code Definition

Code	Description
20	Sale all payment types
26	Void
70	Sale QR Payment
50	Settlement
56	Sale all card scheme payment types
90	Summary Report
91	Detail Report

3.2 Field Type Definition

Code	Description
00	Null Type
01	Approve Code
02	Response Text
03	Transaction Date
04	Transaction Time
06	Merchant Number
16	Terminal Id
30	Primary Account Number / Card Number
31	Expired Date
40	Amount-Transaction
41	Amount-Tip
42	Amount-Cash Back
43	Amount-Tax
44	Amount-Balance
45	Amount-Balance Negative Indicator (Positive/Negative)
50	Batch Number
65	Trace/Invoice Number
D0	Merchant Name and Address
D1	Merchant Id

D2	Card Issuer Name
D3	Reference Number
D4	Card Issuer Id
HA	HOST NAME
HN	NII
HO	Batch Total
E4	APP NAME
D5	Card holder name
D6	Generic Data
D7	Generic Data2
D8	Generic Data3
D9	QR Payment Type
FH	AID
FI	TVR
FJ	TSI
W1	VATB VAT Amount
W2	VATB TAX Allowance
W3	VATB Merchant Unique Value
W4	VATB Campaign Type
W5	VATB Reference 1
W6	VATB Reference 2

A1	QR Type

3.3 Response Code

Code	Description
00 (30 30)	Approve.
CL (43 4C)	Canceled by user operation. (Can be used for health check)
ND (4E 44)	Reject/Error from EDC.
TO (54 4F)	Timeout from host (Host not response).
NN (3n 3n)	Reject from host, the response will be the same as the host return.

3.4 Mapping Issuer Index

Index	Issuer Card Name
01	VISA-CARD
02	MASTERCARD
03	JCB-CARD
04	AMEX-CARD
05	UNIONPAY / K-UNIONPAY
06	TPN-CARD / PROMPT-CARD / VISA-LOCAL
07	KPLUS / PROMPTPAY
08	ALIPAY
09	WECHAT
10	CENTRAL-MC
11	KBank-VISA
12	KBank-MC
13	KBank-JCB / XpressCash-UP
14	QR Credit (QRC_VISA / QRC_MASTERCARD / QRC_UNIONPAY)
15	DOLFIN
16	SCB_IPP

3.5 HOST NAME

Host Name
KBANK
THAI QR
ALIPAY
WECHAT
QR_CREDIT
AMEX
TPN
REDEEM
SMRTPAY
DCC
BAY_INSTALLMENT
AYCAP_T1C
AMEX_EPP
DOLFIN

4 Sale (All Payment Types – Transaction Code 20) Command

4.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	40	12	Amount	
Optional[a,b]	W1	10	VATB VAT Amount	
Optional[a,b]	W2	10	VATB Tax Allowance Amount	
Optional[a,b]	W3	20	Merchant Unique Value	
Optional[a,b]	W4	6	Campaign Type	
Optional[a,c]	W5	20	VATB Reference 1	
Optional[a,c]	W6	20	VATB Reference 2	

Remark:

- Optional[a] means the POS sends all 6 optional fields to the EDC if the government VAT rebate available.
- Optional[b] means the POS sends W1, W2, W3, and W4 to the EDC if the government VAT rebate available.
- Optional[c] means the POS sends W5 and W6 to the EDC if the government VAT rebate available.

4.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response Text from host or EDC	Text will be 40 characters.

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	D0	69	Merchant Name and Address	69 characters, organized as 3 lines of 23 characters.
Mandatory	03	6	Transaction Date	YYMMDD.
Mandatory	04	6	Transaction Time	HHMMSS.
Mandatory	01	6	Approval Code	Host approve code.
Mandatory	65	6	Invoice Number	EDC invoice number.
Mandatory	16	8	Terminal Identification Number (TID)	
Mandatory	D1	15	Merchant Number (MID)	
Mandatory	D2	10	Card Issuer Name	Name of card issuer from EDC.
Mandatory	30	…19	Card Number	Up to 19 digits, format wil be 123456XXXXXX1234 (depend on bank specification).
Mandatory	31	4	Expiry Date	YYMM *Remark: for PCI compliance, the masked digit will be replaced by 'X' (58h).* *Example – 1512 will be replaced by XXXX*
Mandatory	50	6	Batch Number	Batch number from EDC.

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	D3	12	Retrieval Reference Number	Host reference number.
Mandatory	D4	2	Card Issuer ID	Fix value = 12
Mandatory	D5	…27	Card Holder Name	MASK “X” 26 digit
Mandatory	40	12	Amount	Payment amount.

5 Void (All Payment Types – Transaction Code 26) Command

5.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	65	6	Trace or Invoice Number	

5.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response Text from host or EDC	Text will be 40 characters long organized as 2 lines of 20 characters
Mandatory	40	12	Amount	Payment amount.
Mandatory	D0	69	Merchant Name and Address	69 characters, organized as 3 lines of 23 characters
Mandatory	03	6	Transaction Date	YYMMDD
Mandatory	04	6	Transaction Time	HHMMSS
Mandatory	01	6	Approval Code	Host approval code
Mandatory	65	6	Invoice Number	EDC invoice number
Mandatory	16	8	Terminal Identification Number (TID)	
Mandatory	D1	15	Merchant Number (MID)	

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	D2	10	Card Issuer Name	Name of card issuer from EDC tables
Mandatory	30	…19	Card Number	Up to 19 digits. Format will be 123456XXXXXX1234 (depend on bank specification).
Mandatory	31	4	Expiry Date	YYMM * for PCI compliance, the masked digit will be replaced by 'X' (58h). Example.- 1512 will replaced by XXXX
Mandatory	50	6	Batch Number	EDC batch (6 digits long)
Mandatory	D3	12	Retrieval Reference Number	Host reference number
Mandatory	D4	2	Card Issuer ID	Fix value = 12
Mandatory	D5	…27	Card Holder Name	MASK "X" 26 digit

6 Sale QR Payment (Transaction code 70)

6.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	40	12	Amount	
Mandatory	A1	2	QR Type	"01" if Alipay "02" if Wechat "03" if QR Payment "04" if QR Credit

6.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response text from the host or EDC	Text will be 40 characters.
Mandatory	40	12	Amount	Payment amount.
Mandatory	D0	69	Merchant name and address	69 characters, organized as 3 lines of 23 characters.
Mandatory	03	6	Transaction date	YYMMDD.
Mandatory	04	6	Transaction time	HHMMSS.
Mandatory	01	6	Approve code	Host approve code.

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	65	6	Invoice number	EDC invoice number.
Mandatory	16	8	Terminal identification	
Mandatory	D1	15	Merchant number (MID)	
Mandatory	D2	10	Card issuer name	Prompt Pay(K+) = QR PAYMENT ALI PAY = ALIPAY WECHATPAY = WECHANTPAY
Mandatory	30	…64	Primary Account Number / Card Number	Up to 64 digits, format will be 123456XXXXXX1234. The PAN truncate format based on Payment method. Card scheme Payment has PAN truncate. QR Payment doesn't have PAN truncate.
Mandatory	31	4	Expiry date	YYMM *Remark: for PCI compliance, the masked digit will be replaced by 'X' (58h).* *Example – 1512 will be replaced by XXXX*
Mandatory	50	6	Batch number	Batch number from EDC.

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	D3	12	Retrieval Reference	Host reference number.
Mandatory	D4	2	Card Issuer ID	Fix value = 12
Mandatory	D5	…27	Card Holder Name	MASK “X” 26 digit

7 Sale (All Card Scheme Payment Types – Transaction Code 56) Command

7.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	40	12	Amount	
Optional[a,b]	W1	10	VATB VAT Amount	
Optional[a,b]	W2	10	VATB Tax Allowance Amount	
Optional[a,b]	W3	20	Merchant Unique Value	
Optional[a,b]	W4	6	Campaign Type	
Optional[a,c]	W5	20	VATB Reference 1	
Optional[a,c]	W6	20	VATB Reference 2	

Remark:

- Optional[a] means the POS sends all 6 optional fields to the EDC if the government VAT rebate available.
- Optional[b] means the POS sends W1, W2, W3, and W4 to the EDC if the government VAT rebate available.
- Optional[c] means the POS sends W5 and W6 to the EDC if the government VAT rebate available.

7.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response Text from host or EDC	Text will be 40 characters.

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	D0	69	Merchant Name and Address	69 characters, organized as 3 lines of 23 characters.
Mandatory	03	6	Transaction Date	YYMMDD.
Mandatory	04	6	Transaction Time	HHMMSS.
Mandatory	01	6	Approval Code	Host approve code.
Mandatory	65	6	Invoice Number	EDC invoice number.
Mandatory	16	8	Terminal Identification Number (TID)	
Mandatory	D1	15	Merchant Number (MID)	
Mandatory	D2	10	Card Issuer Name	Name of card issuer from EDC.
Mandatory	30	…19	Card Number	Up to 19 digits, format will be 123456XXXXXX1234 (depend on bank specification).
Mandatory	31	4	Expiry Date	YYMM *Remark: for PCI compliance, the masked digit will be replaced by 'X' (58h).* *Example – 1512 will be replaced by XXXX*
Mandatory	50	6	Batch Number	Batch number from EDC.

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	D3	12	Retrieval Reference Number	Host reference number.
Mandatory	D4	2	Card Issuer ID	Fix value = 12
Mandatory	D5	…27	Card Holder Name	MASK “X” 26 digit

8 Settlement (Transaction Code 50) Command each host

8.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	HA	…20	HOST NAME	The host identity. To settle each acquirer one by one, please send the specific HOST NAME as following. Example: KBANK 48 41 00 05 4B 42 41 4E 4B 1. KBANK 2. TPN 3. KPLUS 4. SMARTPAY 5. REDEEM 6. DCC 7. QR_CREDIT 8. ALIPAY 9. WECHAT 10. AMEX 11. AMEX_EPP 12. BAY_INSTALLMENT 13. AYCAP_T1C 14. DOLFIN 15. SCB_IPP 16. SCB_OLS 17. BAY_REDEEM

8.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response text from the host or EDC	Text will be 40 characters.
Mandatory	D0	69	Merchant name and address	69 characters, organized as 3 lines of 23 characters.
Mandatory	16	8	Terminal identification	
Mandatory	D1	15	Merchant number (MID)	
Mandatory	50	6	Batch Number	Batch number from EDC.
Mandatory	03	6	Transaction date	YYMMDD.
Mandatory	04	6	Transaction time	HHMMSS.
Mandatory	HN	3	NII	
Mandatory	HO	96	Batch Totals	Capture Sale Count 3 Capture Sale Amount 12 Capture Refund Count 3 Capture Refund Amount 12 Debit Sale Count 3 Debit Sale Amount 12 Debit Refund Count 3 Debit Refund Amount 12 Authorize Sale Count 3

Mandatory / Optional	Field	Length	Field Name	Comments
				Authorize Sale Amount 12 Authorize Refund Count 3 Authorize Refund Amount 12

9 Settlement (Transaction Code 50) Command ALL HOST

9.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	HN	3	NII	To settle all acquirers within one command, please send the specific NII = 999.

9.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	ZY		Host total	Size : 128 bytes per hosts Data: concatenated string in format H%03uM%03u%s H <3 bytes host index>M<3 bytes merchant no><SUMMARY (ISO8583 DE63)>
Mandatory	ZZ	XX	Batch Settle Status	Host KBANK = 000120NA Host DCC = 001121NA Host Smart Pay = 002120NA Host Redeem = 003120NA Host TPN = 004125NA Host AMEX = 005003NA Host K+ = 00613000 Host We Chat = 007130NA Host Ali Pay = 00813000 Host QR Credit = 009130ER *** Example*** 000 = Host INDEX 120 = NII

Mandatory / Optional	Field	Length	Field Name	Comments
				NA = Settlement Status Skip Host = NA Success = 00 Unsuccess = not "00" ER – error from EDC. any number - Response from host.

10 SUMMARY Report (Transaction code 90) each host

10.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	HA	…20	HOST NAME	The host identity. To settle each acquirer one by one, please send the specific HOST NAME as following. Example: KBANK 48 41 00 05 4B 42 41 4E 4B 1. KBANK 2. TPN 3. KPLUS 4. SMARTPAY 5. REDEEM 6. DCC 7. QR_CREDIT 8. ALIPAY 9. WECHAT 10. AMEX 11. AMEX_EPP 12. BAY_INSTALLMENT 13. AYCAP_T1C 14. DOLFIN 15. SCB_IPP 16. SCB_OLS 17. BAY_REDEEM

10.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response text from the host or EDC	Text will be 40 characters.
Mandatory	D0	69	Merchant name and address	69 characters, organized as 3 lines of 23 characters.
Mandatory	16	8	Terminal identification	
Mandatory	D1	15	Merchant number (MID)	
Mandatory	50	6	Batch Number	Batch number from EDC.
Mandatory	03	6	Transaction date	YYMMDD.
Mandatory	04	6	Transaction time	HHMMSS.
Mandatory	HN	3	NII	
Mandatory	HO	96	Batch Totals	Capture Sale Count 3 Capture Sale Amount 12 Capture Refund Count 3 Capture Refund Amount 12 Debit Sale Count 3 Debit Sale Amount 12 Debit Refund Count 3 Debit Refund Amount 12 Authorize Sale Count 3 Authorize Sale Amount 12 Authorize Refund Count 3 Authorize Refund Amount 12

11 SUMMARY Report (Transaction code 90) all host

11.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	HN	3	NII	The host identity. To settle all acquirer, please send the NII "999".

11.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response text from the host or EDC	Text will be 40 characters.
Mandatory	D0	69	Merchant name and address	69 characters, organized as 3 lines of 23 characters.
Mandatory	16	8	Terminal identification	Fix data "00000000"
Mandatory	D1	15	Merchant number (MID)	Fix data "000000000000000"
Mandatory	50	6	Batch Number	Fix data "000001"
Mandatory	03	6	Transaction date	YYMMDD.
Mandatory	04	6	Transaction time	HHMMSS.
Mandatory	HN	3	NII	Fix data "999"
Mandatory	HO	96	Batch Totals	Capture Sale Count 3 Capture Sale Amount 12 Capture Refund Count 3 Capture Refund Amount 12 Debit Sale Count 3 Debit Sale Amount 12

Mandatory / Optional	Field	Length	Field Name	Comments
				Debit Refund Count 3 Debit Refund Amount 12 Authorize Sale Count 3 Authorize Sale Amount 12 Authorize Refund Count 3 Authorize Refund Amount 12

12 Detail Report (Transaction code 91) command each host

12.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	HA	…20	HOST NAME	The host identity. To settle each acquirer one by one, please send the specific HOST NAME as following. Example: KBANK 48 41 00 05 4B 42 41 4E 4B 1. KBANK 2. TPN 3. KPLUS 4. SMARTPAY 5. REDEEM 6. DCC 7. QR_CREDIT 8. ALIPAY 9. WECHAT 10. AMEX 11. AMEX_EPP 12. BAY_INSTALLMENT 13. AYCAP_T1C 14. DOLFIN 15. SCB_IPP 16. SCB_OLS 17. BAY_REDEEM

12.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response text from the host or EDC	Text will be 40 characters.
Mandatory	D0	69	Merchant name and address	69 characters, organized as 3 lines of 23 characters.
Mandatory	16	8	Terminal identification	
Mandatory	D1	15	Merchant number (MID)	
Mandatory	50	6	Batch Number	Batch number from EDC.
Mandatory	03	6	Transaction date	YYMMDD.
Mandatory	04	6	Transaction time	HHMMSS.
Mandatory	HN	3	NII	
Mandatory	HO	96	Batch Totals	Capture Sale Count 3 Capture Sale Amount 12 Capture Refund Count 3 Capture Refund Amount 12 Debit Sale Count 3 Debit Sale Amount 12 Debit Refund Count 3 Debit Refund Amount 12 Authorize Sale Count 3 Authorize Sale Amount 12

Mandatory / Optional	Field	Length	Field Name	Comments
				Authorize Refund Count 3 Authorize Refund Amount 12

13 Detail Report (Transaction code 91) all host

13.1 Request from POS to EDC

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	HN	3	NII	The host identity. To settle all acquirer, please send the NII "999".

13.2 Response from EDC to POS

Mandatory / Optional	Field	Length	Field Name	Comments
Mandatory	02	40	Response text from the host or EDC	Text will be 40 characters.
Mandatory	D0	69	Merchant name and address	69 characters, organized as 3 lines of 23 characters.
Mandatory	16	8	Terminal identification	Fix data "00000000"
Mandatory	D1	15	Merchant number (MID)	Fix data "000000000000000"
Mandatory	50	6	Batch Number	Fix data "000001"
Mandatory	03	6	Transaction date	YYMMDD.
Mandatory	04	6	Transaction time	HHMMSS.
Mandatory	HN	3	NII	Fix data "999"
Mandatory	HO	96	Batch Totals	Capture Sale Count 3 Capture Sale Amount 12 Capture Refund Count 3 Capture Refund Amount 12 Debit Sale Count 3 Debit Sale Amount 12

Mandatory / Optional	Field	Length	Field Name	Comments
				Debit Refund Count 3 Debit Refund Amount 12 Authorize Sale Count 3 Authorize Sale Amount 12 Authorize Refund Count 3 Authorize Refund Amount 12

Appendix A. VAT REBATE

สินค้า	ยอดเงิน	ยอดเงิน	VAT
ปากกา	100		
ขนมไส้ทุเรียน	100	100 OTOP	7
หนังสือ	100	100 Book	7
เหล้า	100		
รวม	400		
VAT 7%	28	200 (Amount for TAX Allowance)	14 (VAT Rebate)
ยอดที่จ่าย	428		

วิธีการ และ Field สำหรับใช้ส่งข้อมูล VAT - 61

- ธนาคารสมาชิกพิจารณาเลือก Field ที่จะใช้ส่งข้อมูล VAT, Amount of VAT, Merchant Unique Value , Campaign Type ร่วมกัน
- ตัวอย่าง 00000000000000140000000000000002000000010293000000000257190001

Field No.	Type	RQ	RP	Field Description
61	1 B + up to 62 AN	C		**VAT information**
	Subfield	**Position**	**Format**	**Field Description**
	1	1	n-18	**VAT Rebate amount** In authorization requests, field 61.1 contains the cashback amount for welfare card and debit card. This field is used in 0100 authorization requests, 0400 and 0420 reversal requests, not support to 0110 authorization responses, 0410 and 0430 reversal responses. Example: 14.00 Baht is 000000000000001400
	2	19	n-18	**Amount for TAX allowance** จำนวนเงินที่นำมาคำนวณ VAT ใน 61.1 Example: 200.00 Baht is 000000000000020000
	3	37	an-20	**Merchant Unique Value** ข้อมูลอ้างอิงที่สร้างจาก POS ที่เป็น Unique ของแต่ละรายการ เพื่อใช้ในกรณีตรวจสอบข้อมูลย้อนกลับ Example: 00010293000000000257
	4	57	an-6	**Campaign Type** หมายเลขโครงการ YYNNNN เมื่อ YY ค.ศ. NNNN คือ Numbers Example: 190001 มาตรการภาษีเพื่อกระตุ้นการซื้อสินค้าเกี่ยวกับการศึกษา 190002 มาตรการภาษีเพื่อส่งเสริมสินค้าท้องถิ่นไทย (OTOP)